

Advanced Programming

Introduction to Java language



Java language

- The 1st version was made available in 1995 by Sun Microsystems
 - Its study started in 1991
- Fully object-oriented language
- Cross-platform
- The syntax follows the C/C++ style



Creation of a Java program

- Just like in other languages, creating a Java application involves writing the code, using...
 - Traditional text editor
 - notepad, notepad++, ...
 - vi, emacs, nano, ...
 - TextEdit, TextWrangler, BBEdit, ...
 - Visual Studio Code
 - ...
 - Integrated Development Environment (IDE)
 - NetBeans
 - Eclipse
 - **IntelliJ (free license available for students)**
 - **Visual Studio Code (with Microsoft Extension Pack for Java)**
 - ...

Compiling a Java program

- Compilation, which includes syntax checking and generation of compiled code, is performed using the *Java Development Kit* (JDK)
 - Some development environments have built-in versions
 - Includes tools and class libraries
 - Includes a compilation tool: `javac`
 - Intermediate code, called *Java bytecode*, is generated
 - Architecture independent code

JDK installation

- *OracleJDK vs OpenJDK*



- *OracleJDK*

- Commercial version, requiring licensing
- Support provided by *Oracle*
- Supposedly more stable and with a better performance

- *OpenJDK*

- *Open source* (we can contribute to its development)
- Although its support is primarily done by its active development communities, there are specific distributions with their own support

JDK installation

- JDK version to install: Java SE
 - The latest available version is 23
 - *"JDK 23 will receive updates under these terms, until March 2025, when it will be superseded by JDK 24."*
- Available at:
 - OracleJDK
 - <https://www.oracle.com/java/technologies/downloads/>
 - OpenJDK
 - <https://jdk.java.net/23/>
- Version 21 is a *Long-Term Support* (LTS) release and will be supported until September 2026

JDK installation

- Unzip the downloaded JDK archive to an appropriate folder
 - **Windows:** C:\Program Files\Java
 - **macOS:** /Library/Java/JavaVirtualMachines
- Set the `JAVA_HOME` environment variable to the base folder of the JDK
 - **Windows:** Use the "System Properties" > "Advanced" > "Environment Variables"
 - **macOS:** Add the following line to `.zshrc`, `.zshenv`, or `.zprofile`:

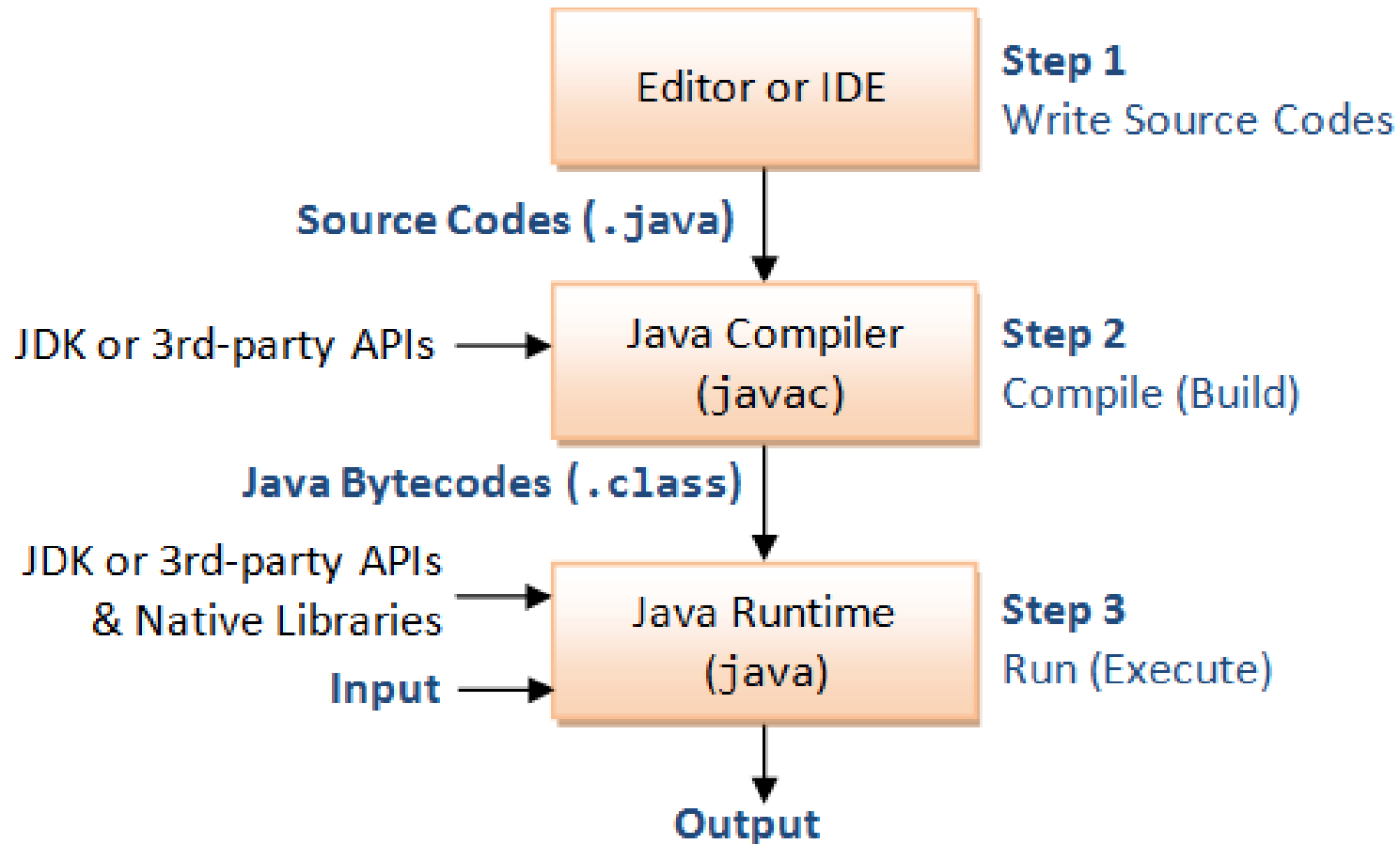
```
export JAVA_HOME=/Library/Java/JavaVirtualMachines/jdk-23.0.2.jdk/Contents/Home
```
- Add the `bin` folder of the JDK to the `PATH` environment variable
 - **Windows:** Use the "System Properties" > "Advanced" > "Environment Variables"
 - **macOS:** Add the following line to `.zshrc`, `.zshenv`, or `.zprofile`:

```
export PATH=$JAVA_HOME/bin:$PATH
```

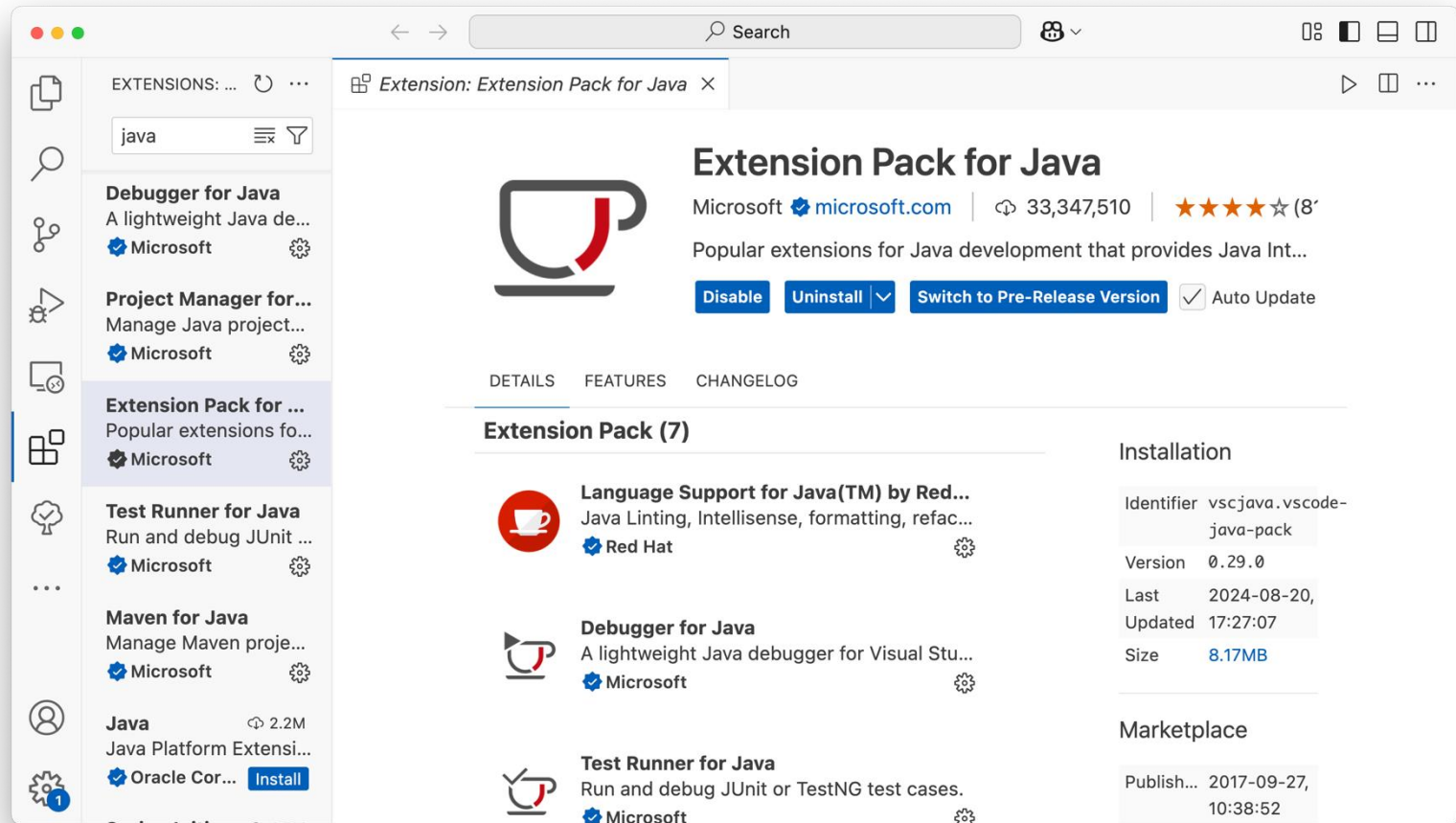
Java Virtual Machine

- As the compilation result is not native code, it is necessary to have a *Java bytecode interpreter*
- The *Java bytecode interpreter* is called *Java Virtual Machine* (JVM)
 - The JDK includes the "java" command to run programs previously compiled to *Java bytecode* (.class files)
 - The JVM provides a secure, isolated environment in which programs run without affecting or being affected by other programs (*sandbox*)

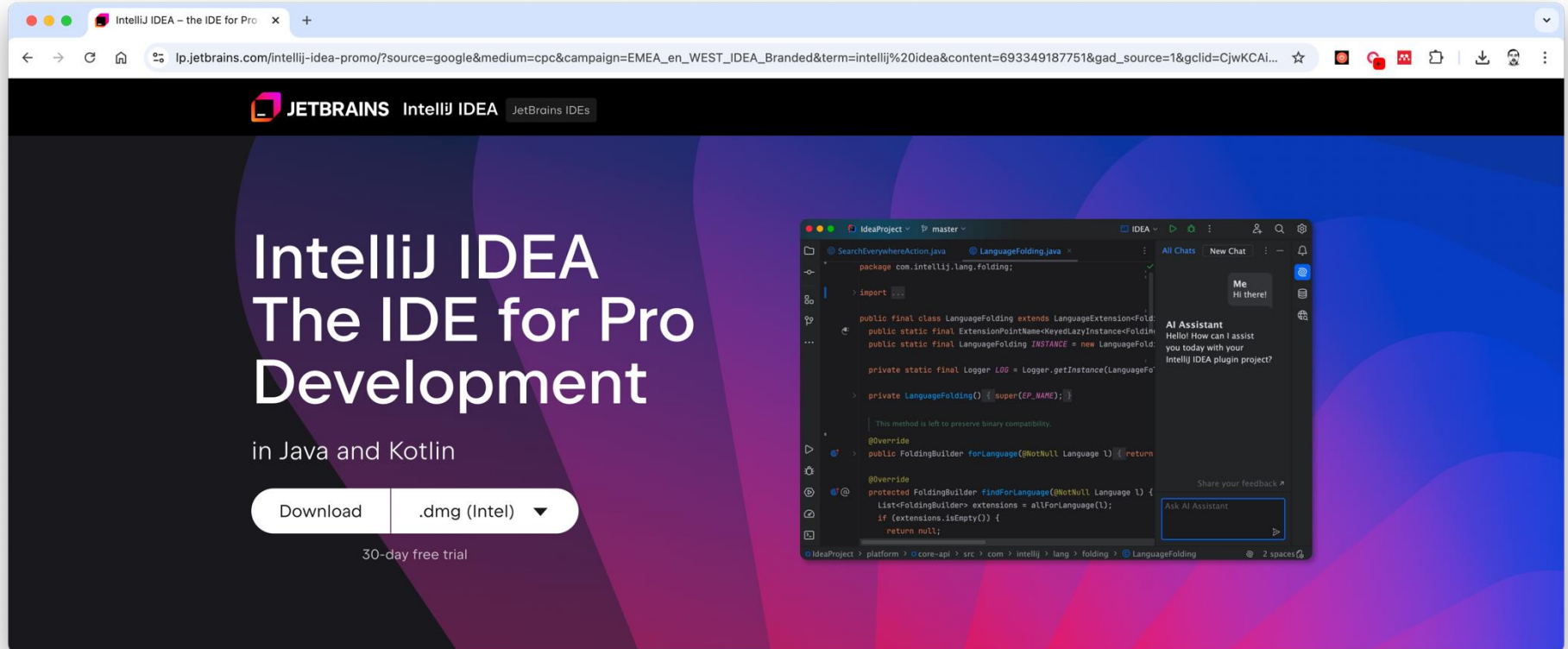
Creation and execution



Visual Studio Code – Extension Pack for Java



IntelliJ – JetBrains



The screenshot shows the IntelliJ IDEA website in a web browser. The browser's address bar displays the URL: `lp.jetbrains.com/intellij-idea-promo/?source=google&medium=cpc&campaign=EMEA_en_WEST_IDEA_Brand&term=intellij%20idea&content=693349187751&gad_source=1&gclid=CjwKCAI...`. The website header features the JetBrains logo and the text "IntelliJ IDEA" and "JetBrains IDEs". The main content area has a dark background with a blue and purple abstract design. It prominently displays "IntelliJ IDEA" and "The IDE for Pro Development" in large white text, followed by "in Java and Kotlin". Below this, there are two buttons: "Download" and ".dmg (Intel)" with a dropdown arrow. A "30-day free trial" tag is positioned below the ".dmg (Intel)" button. On the right side of the website, there is an inset image of the IntelliJ IDEA IDE interface. This inset shows a code editor with a Java file named `LanguageFolding.java`. The code includes package declarations, imports, and class definitions. A chat window titled "AI Assistant" is open on the right side of the IDE, displaying a conversation where the user says "Hi there!" and the assistant responds with a greeting and an offer to assist with the IntelliJ IDEA plugin project. The IDE interface also shows a sidebar with project files and a bottom status bar.

IntelliJ IDEA
The IDE for Pro Development
in Java and Kotlin

Download .dmg (Intel) ▼

30-day free trial

```
package com.intellij.lang.folding;

import ...

public final class LanguageFolding extends LanguageExtension<Fold...
public static final ExtensionPointName<KeyedLazyInstance<Foldin...
public static final LanguageFolding INSTANCE = new LanguageFold...

private static final Logger LOG = Logger.getInstance(LanguageFo...

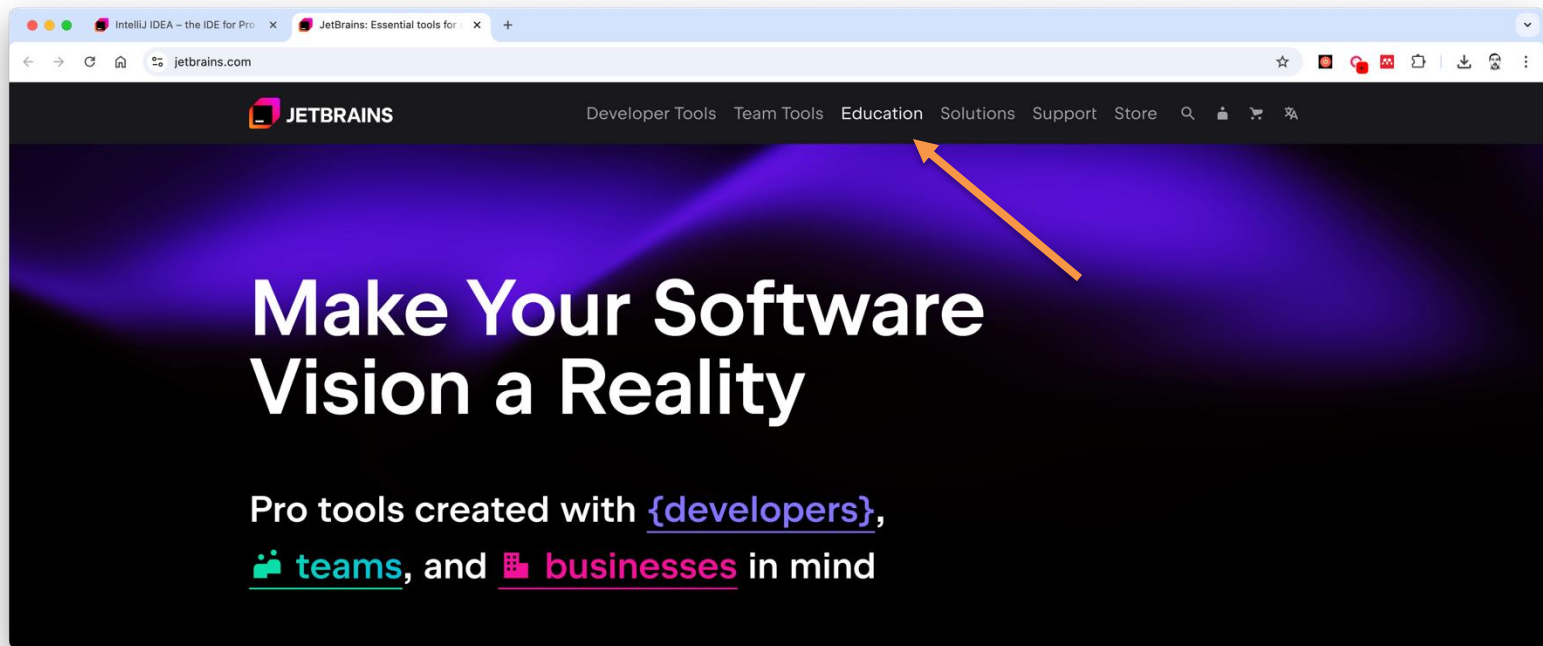
private LanguageFolding() { super(EP_NAME); }

...

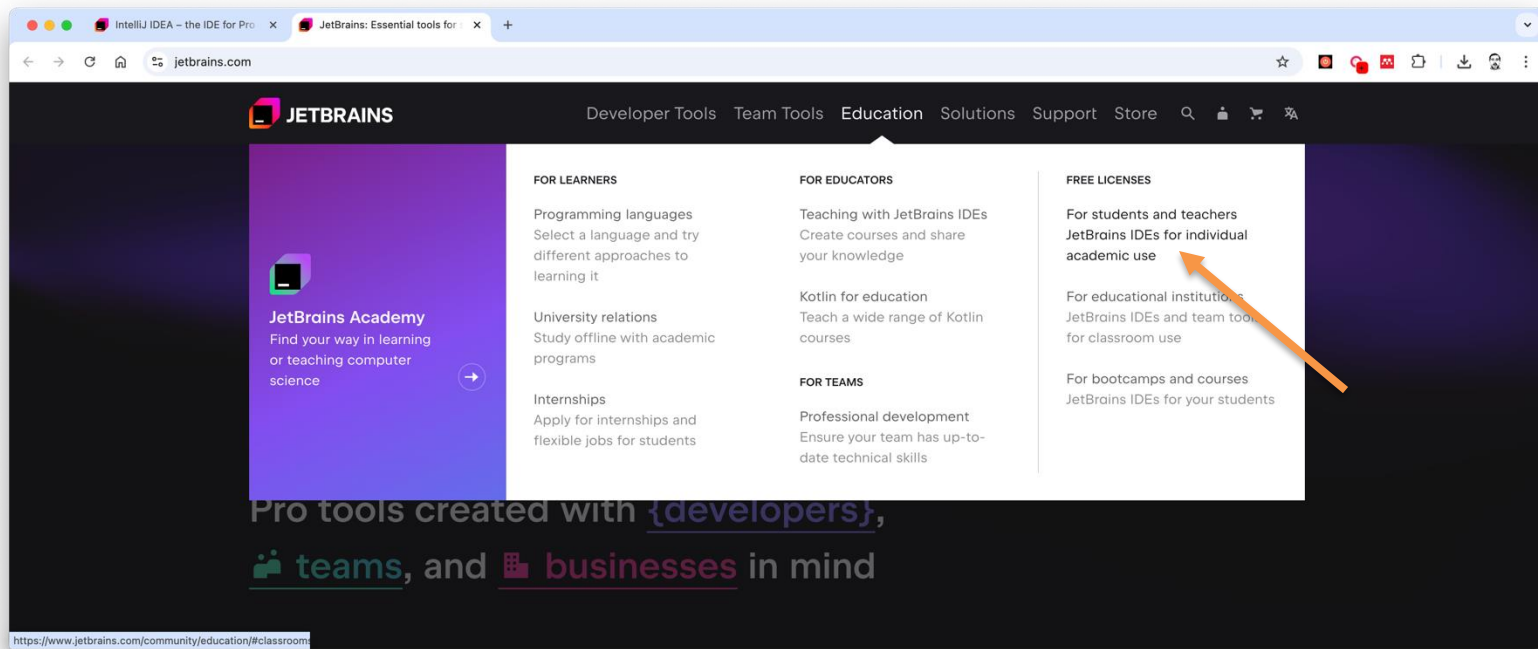
@Override
public FoldingBuilder forLanguage(@NotNull Language l) { return...

@Override
protected FoldingBuilder findForLanguage(@NotNull Language l) {
    List<FoldingBuilder> extensions = allForLanguage(l);
    if (extensions.isEmpty()) {
        return null;
    }
}
```

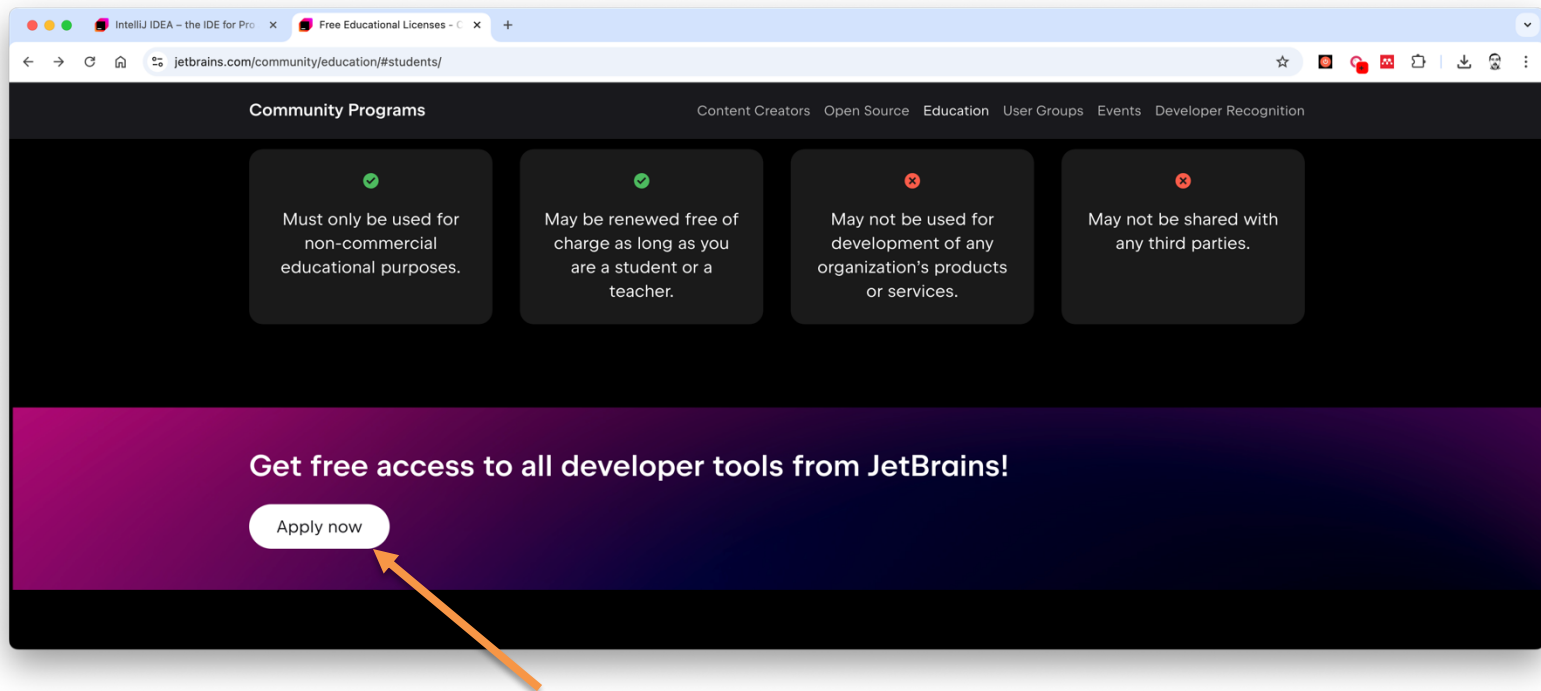
- How to get the educational license to use JetBrains tools?



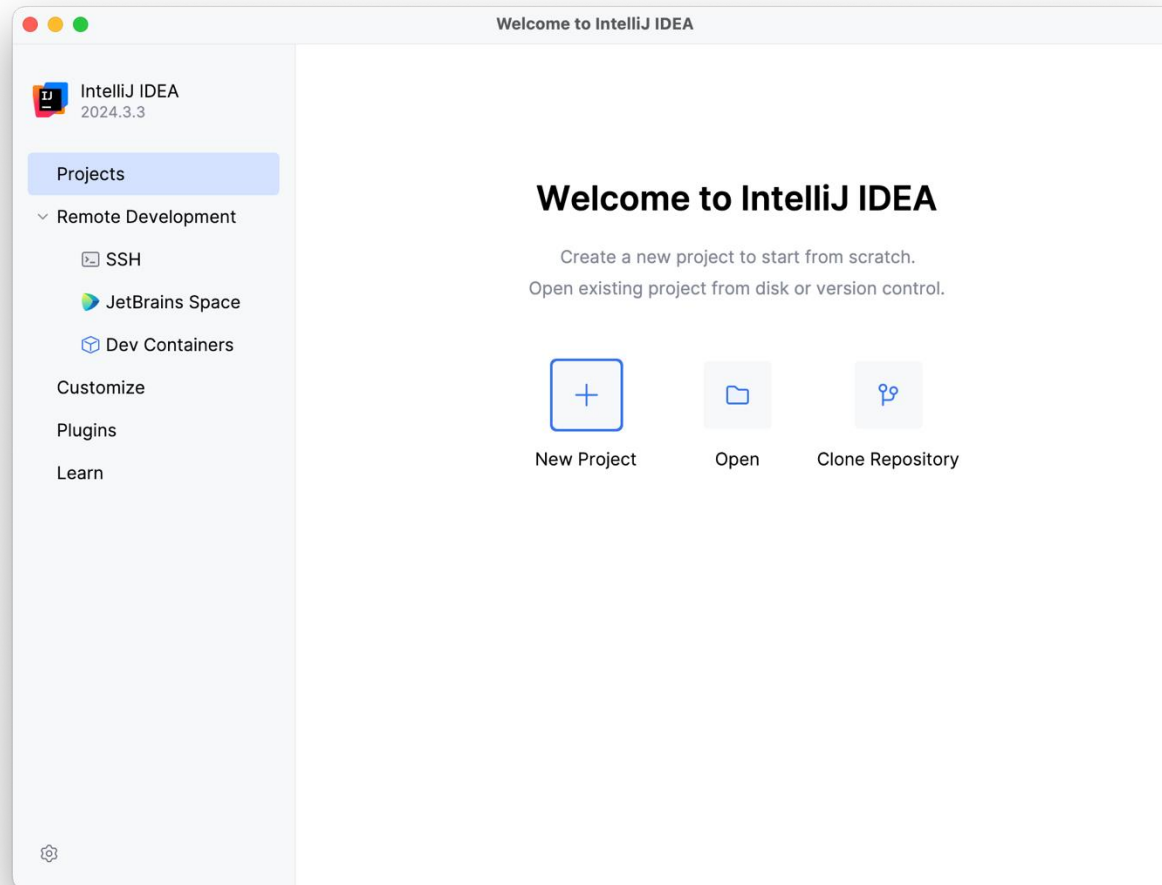
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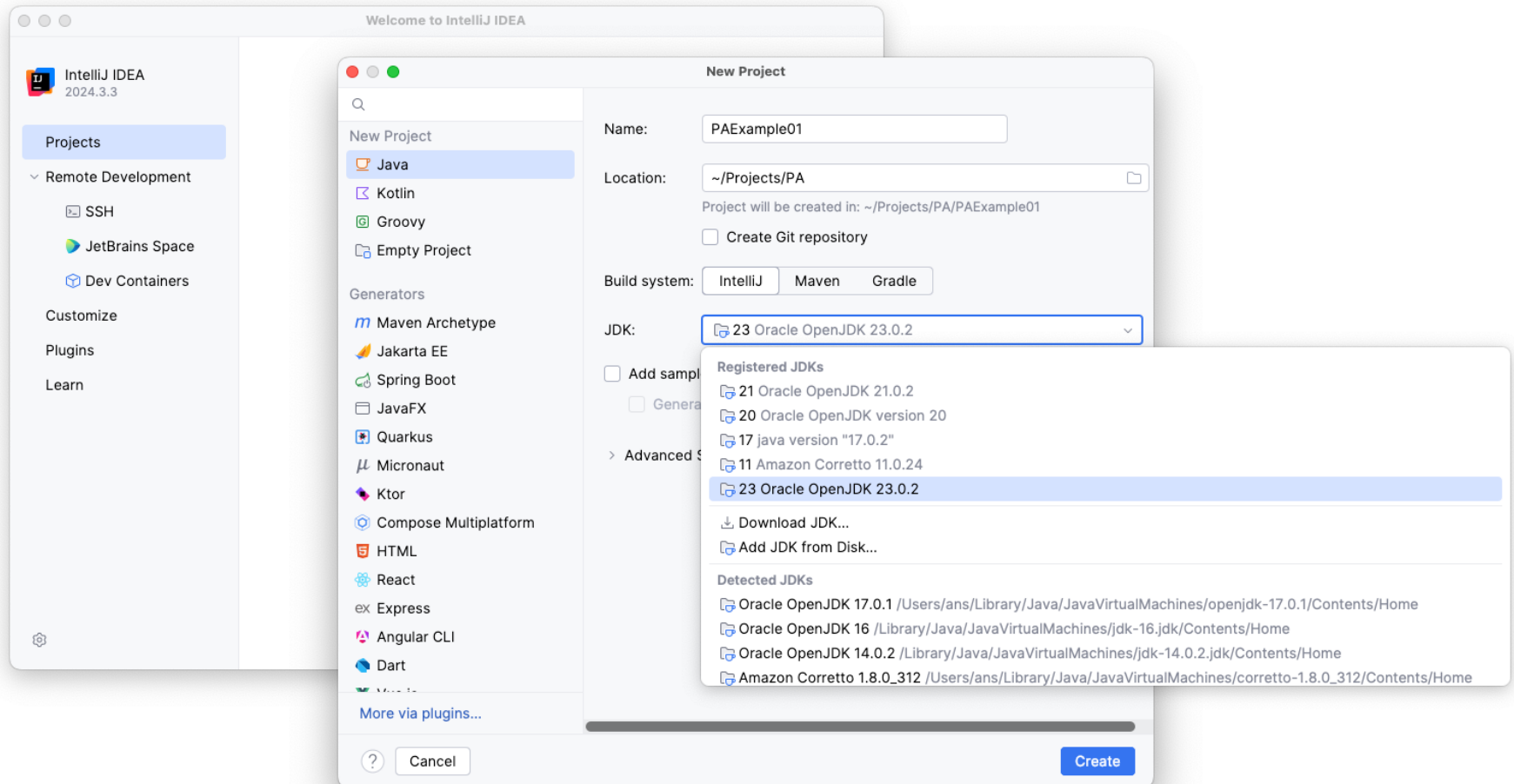


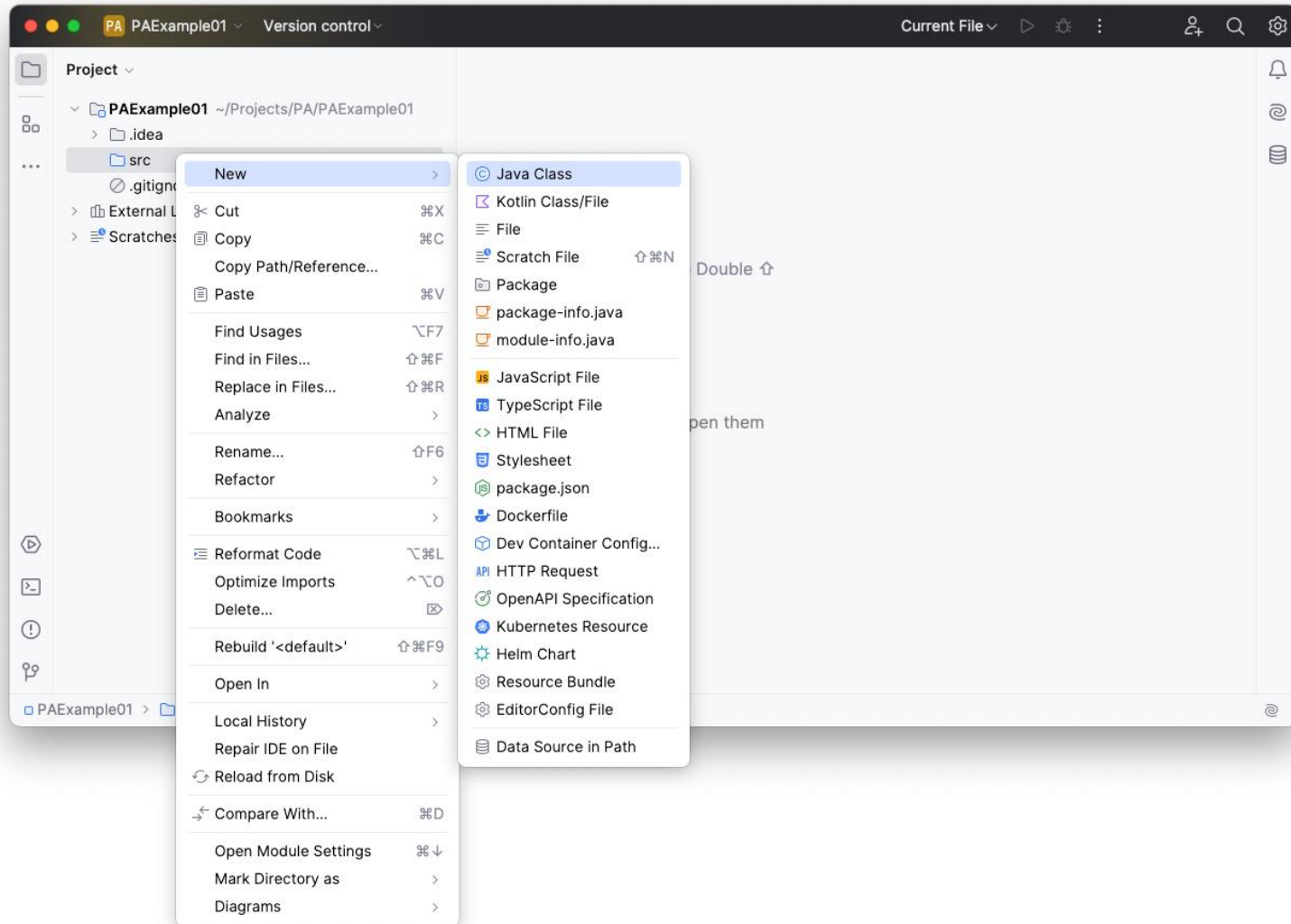
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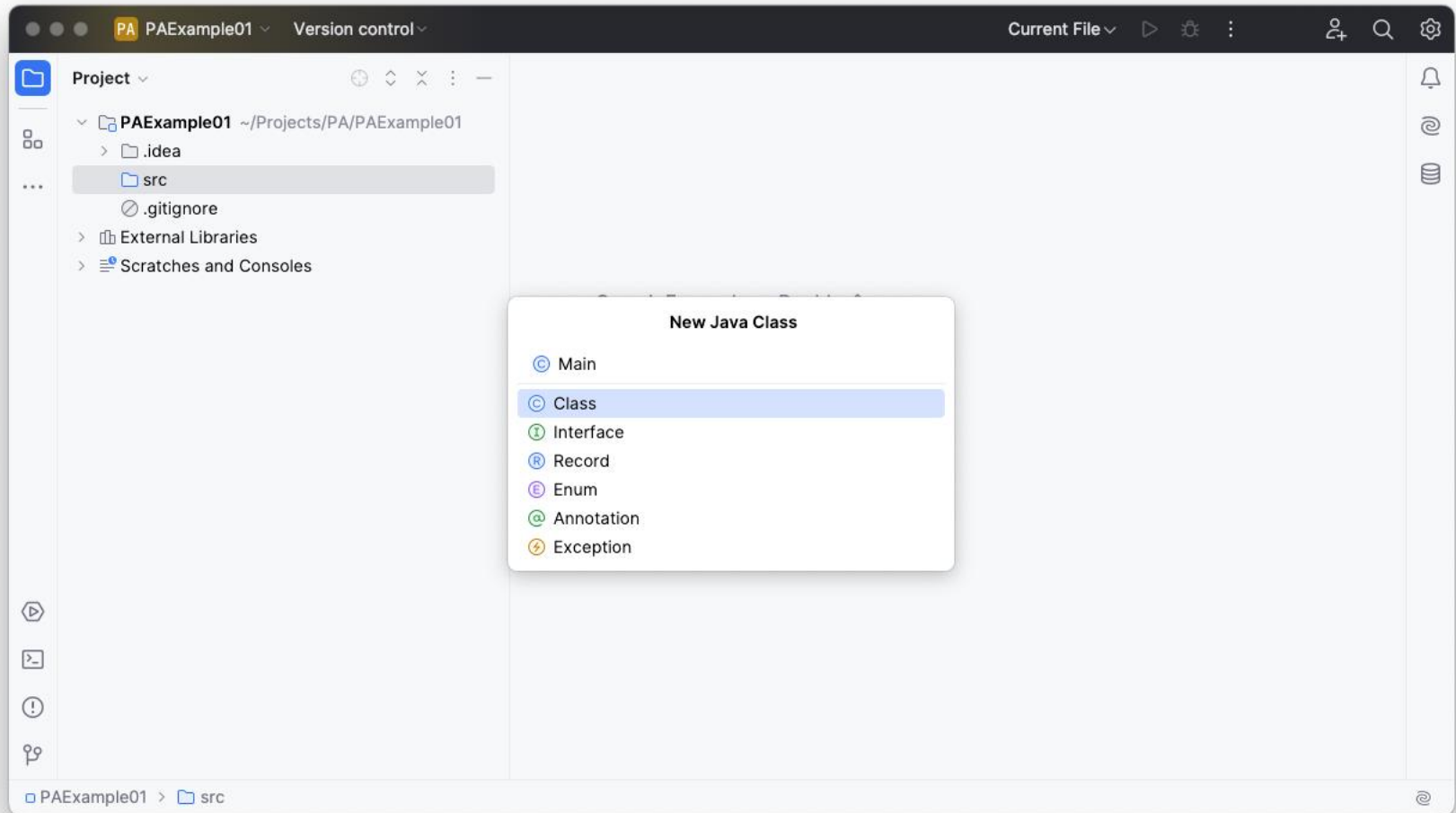


- Create a new project





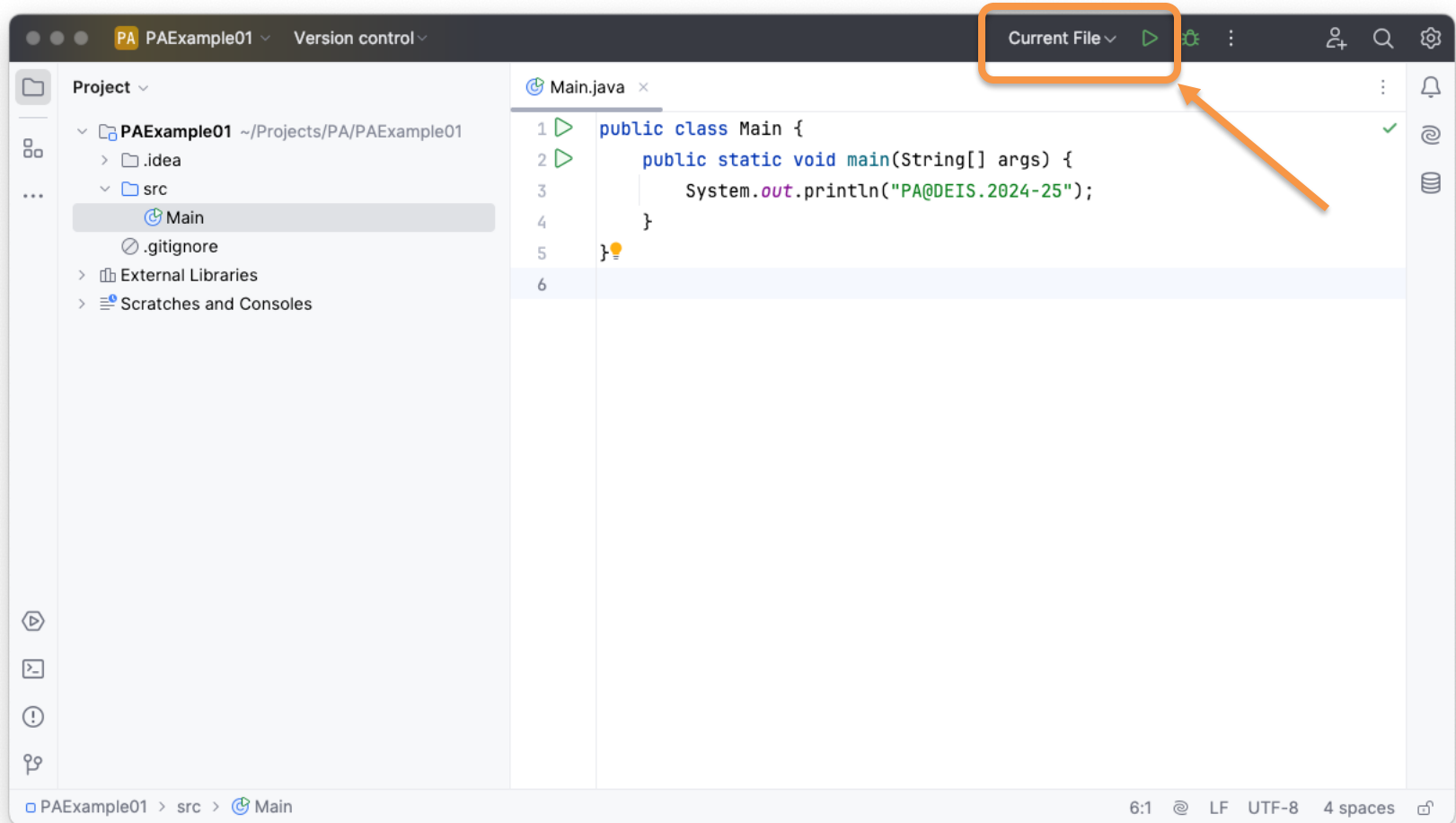




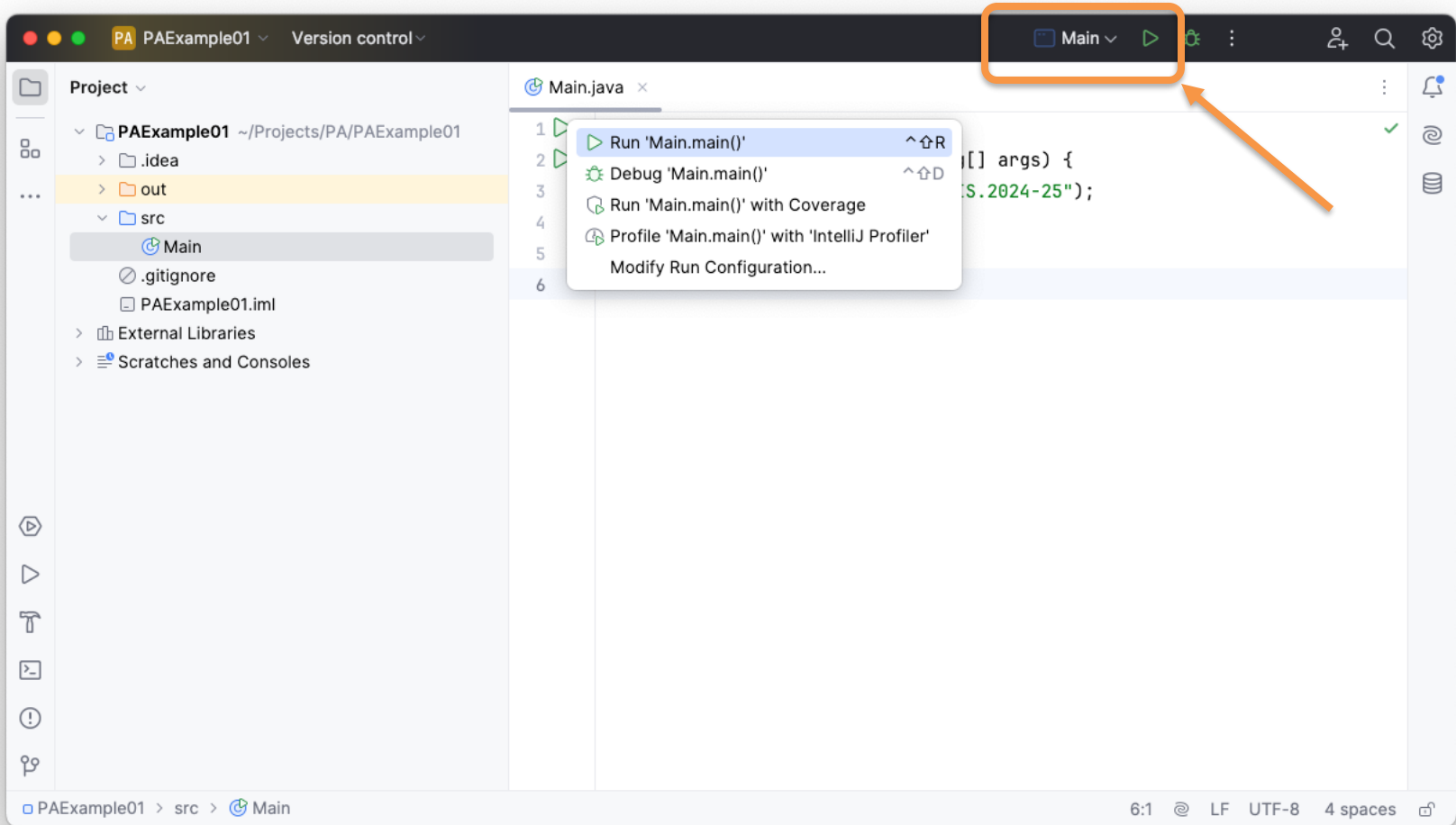
Example

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("PA@DEIS.2024-25");  
    }  
}
```

Running the example



Running the example – Run Configurations



Advanced Programming

Exercises

Exercise

- Exercise 1

- Write an application that generates a random integer between 0 and 100 (inclusive)
- The user must guess the generated number by entering successive values
- The program should keep informing the user whether the guessed number is higher or lower than the generated number

Exercise

- Exercise 2

- Write an application that asks the user to think of a number between 1 and 100, inclusive
- By asking the user a series of questions, the application will try to guess the number
- The user will respond with 'CORRECT', 'MINE IS LOWER', or 'MINE IS HIGHER'
- At the end, the application should display the number of attempts needed to guess correctly

Exercise

• Exercise 3

- Write an application that creates an array of integers by asking the user for the array's size and the values of its elements
- Based on the created array, develop static functions to:
 - Print the array (numbers separated by spaces)
 - Calculate and print the largest and smallest elements in the array
 - Sum all the elements of the array, returning the sum
 - Calculate and print the average of the values in the array
 - Reverse all the values in the array, $[i_0, i_1, \dots, i_{n-1}] \Rightarrow [i_{n-1}, \dots, i_1, i_0]$
 - Retrieve all elements that are greater than the average, using only the given array and without calling any other function